Intermediate robotic T. rex

A 60 cm simulation platform for indoor walking, obstacle avoidance and operator-selected person following.



R3 / 175 mm sole spacing / nominal standing pose

1.10 m
nose-to-tail envelope

10.37 kg
allocated carried mass

$16,447
prototype target / high $21,817

18.4 min
estimated at 179.5 W, 75% usable

Why 60 cm

At the same height, the redesigned robot fills 84% of a fixed side-view window around the hips, up from 48%. Curved chest, jaw and tail panels hide the electronics while keeping the leg drives accessible. The 75 and 90 cm designs give more space, but demand more lateral support torque.

Locomotion

Original R3 stayed upright in 20/20 trials at a median 0.114 m/s. All missed the smoothness target. With a revised action interface, raw PPO passed 15/20 courses versus 1/20 for its reference. On fresh seeds it passed 13/20, versus 2/20 for the reference and 3/20 at the previously selected gain.

Person following

The original hybrid resumed following after a cart crossed in 4/4 trials and passed 1/4 initial pedestal detours. On two fresh seeds and both floors, raw PPO passed 3/4 detours, the 0.35-gain hybrid 2/4 and the zero-PPO reference 3/4. The smoother actor passed 1/4 cart and 0/4 pedestal trials.

Proximal policy optimization (PPO) trains a neural controller to adjust the reference gait.

From simulator to test fixture

R3 brings the feet closer and mounts the fixed arms to the chassis rails. In single support, one hip needs 9 N·m against a 6 N·m continuous allowance. Test the drive and bearings under load and measure motor temperature, battery current, traction and panel clearance in a catch frame before sizing the 3–4 ft robot.